

16. Let p , q and r be the propositions
 p : You get an A on the final exam q : You do every exercise in this book.
 r : You get an A in this class.

Write these propositions using p , q and r and logical connectives (including negations).

- a) You get an A in this class, but you do not do every exercise in this book — $r \wedge \neg q$.
- b) You get an A on the final, you do every exercise in this book, and you get an A in this class — $p \wedge q \wedge r$.
- c) To get an A in this class, it is necessary for you to get an A on the final — $r \rightarrow p$.
- d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class — $(p \wedge \neg q) \vee r$?
- e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class — $(p \wedge q) \rightarrow r$.
- f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final — $(r \leftrightarrow (q \vee p))$

17. Let p , q and r be the propositions
 p : Grizzly bears have ^{been} seen in the area.
 q : Hiking is safe on the trail.
 r : Berries are ripe along the trail.

Write these propositions using p , q and r and logical connectives (including negations).

- a) Berries are ripe along the trail, but grizzly bears have not been seen in the area — $r \wedge \neg p$ ✓
- b) Grizzly bears have not been seen in the area and hiking on the trail is safe, but berries are ripe along the trail — $(\neg p \wedge q) \wedge r$ ✓
- c) If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area — $r \rightarrow (q \leftrightarrow \neg p)$ ✓